

Suryansh Aryan

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EDUCATION

Kevin T. Crofton Department of Aerospace Engineering, Virginia Tech Blacksburg, VA, USA
M.S Thesis in Aerospace Engineering, Specialization in Dynamics & Control, CGPA: 3.85/4.00 Aug 2023 – May 2026

Manipal University Manipal, KN, India
Bachelors of Technology in Aerospace Engineering, CGPA: 9.09/10.00 July 2019 – July 2023

EXPERIENCES

Graduate Research Assistant May 2024 – Present
Center for Space Science and Engineering, Virginia Tech Blacksburg, VA, USA

- Spearheaded NSF-funded project development as part of space engineering team, with 2 other teams from NC State University & University of Surrey, driving research solutions in near-Earth communication resiliency.
- Development and maintainance of Virginia Tech's in-house deep space emulator, SpaceNet, first of its kind MiniNet based network topology generator outperforming state-of-the-art network emulators like Hypatia, NS3 in accuracy and computational time
- Led development of realistic user behaviors and satellite distributions and published work at Aerospace conferences at SciTech 2025 & SmallSat 2025

UAV Research Intern Jan. 2023 – July 2023
Computational Intelligence Lab, Indian Institute of Science (IISc) Bengaluru, India

- Evaluated 4 key challenges to sustainable precision agriculture via drone technology & constructed a costeffective architecture improving efficiency & affordability for Indian farmers upto \$3000.
- Achieved over \$2,000 cost savings on high-end electronics & reduced maintenance expenses by up to two years.
- Built GNC modules with SPI & CAN-based radars & depth camera drivers for PX4 & K++ flight controllers.
- Improved canopy coverage efficiency by 40%, optimizing variable spraying for crops and trees through ROS/Gazebo simulations.

Guidance-Navigation-Control Intern May 2022 – Aug 2022
General Aeronautics Pvt Ltd Bengaluru, India

- Spearheaded the development of scalable SITL & HITL package enhancing trajectory planning for the company's autonomy project, saving \$1000+ on expensive flight controllers & high-resolution sensors.
- Developed 3 innovative algorithms enabling decision-making to detect and navigate obstacles and targets, achieving precision agriculture objectives, saving 3+ months of the control division's efforts.
- Implemented a robust metaheuristic GNC algorithm with APF collision avoidance in SITL, enhancing efficiency by 20%, saving 100+ hours of ground testing.

PROJECTS

Fault Tolerant Lunar C&PNT Architecture | *Python, MATLAB, STK 12, Mininet* Jan 2025 – Apr 2026

- Developed an in-house Lunar Delay Tolerant Network (DTN) emulator for sustainable communication & PNT service studies of ELFO type C&PNT service constellation.
- Built navigation infrastructure with least square & EKF solutions for moving users and communication infrastructure in compliance with LunaNet protocol standards on BPv7 protocol & TCP convergence layer.

Distributed Fault-tolerant Asteroid Inspection | *MATLAB, Simulink, Spice*

- Constructed an integrated Adhoc SMC controller with stochastic MPC controller to incorporate unmatched uncertainties in asteroid dynamics under frozen orbit conditions for distributed asteroid inspection problem.
- Built a swarm reconfiguration strategy that generates optimal satellite-target pairs and compute desired reference trajectories for the adaptive controller using collocation-based BVP formulation.

Four Body Invariant Structures For Jovian System | *MATLAB, STK* Aug 2023 – Aug 2024

- Researched families of quasi-periodic invariant tori around the liberation points of Europa and Callisto and found natural transfers due to Ganymede's periodic assist in a Planar Four Body Problem (PCCFBP).
- Implemented stroboscopic mapping on Gauss-Legendre collocation method to determine families of invariant tori in 2D phase-space
- Presented the research findings with Dr. Riley Fitzgerald under "Orbital Dynamics, Perturbations & Stability" session at AAS Astrodynamics Specialist Conference 2024, Broomfield, CO, USA.

1. **Aryan S.** et. al., LEO Satellite Network Testbed: MiniNet-based Emulation of Network Congestion and Routing Dynamics. *Acta Astronautica*. (In-Progress)
2. B. Barbour, Kedrowitsch A., Downs J., and Mhadgut D., **Aryan S.**, Black J., Kenyon S. "SpaceNet: A Resource-Efficient Simulation Testbed for LEO Mega-Constellation Networks," in *IEEE Access*, vol. 14, pp. 28097-28118, 2026, doi: 10.1109/ACCESS.2026.3666138. (IEEE Journal)
3. **Aryan, S.**, Downs J., Kedrowitsch A., Smith J., Houck A. and, Kenyon S. SpaceNet Testbed – Small Satellite Network Emulation of Real-World Scenarios. *DigitalCommons@USU - Small Satellite Conference 2025*, Salt Lake City, Utah. (Poster presentation - 3rd best student poster award)
4. **Aryan, S.** Reference tracking of nonlinear airborne systems using stochastic MPC with disturbance observers and actuator chance constraint optimization. *CEAS Aeronaut J* (2025). <https://doi.org/10.1007/s13272-025-00847-w> (Springer Nature journal)
5. Downs, J. & Barbour, B. & Kedrowitsch, A. & Mhadgut, D. & **Aryan, S.** & Kenyon, S. Space Network (SpaceNet) Testbed - Development of a Multi-Functional Testbed for Simulating Space Communication Networks. *AIAA Scitech 2025*. 10.2514/6.2025-2716. (Conference proceedings)
6. **Aryan, S.** & Fitzgerald, R. Four Body Invariant Structures And Chaos Analysis For Jovian Multi-Moon Ballistic Transfers. *AAS Astrodynamics Specialist conference 2024, Broomfield, CO*. (Conference proceedings)
7. More, D., Acharya, S., and **Aryan, S.**, "SRGAN-TQT, an Improved Motion Tracking Technique for UAVs with Super-Resolution Generative Adversarial Network (SRGAN) and Temporal Quad-Tree (TQT)," *SAE Technical Paper 2022-26-0021*, 2022, <https://doi.org/10.4271/2022-26-0021>. (Conference proceedings)

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Julia, Markdown, HTML/CSS

Software Tools: MATLAB/Simulink, LabView, Mathematica, Robot Operating System (ROS), Ansys STK, GMAT, KSP, FlightGear, QGC, MissionPlanner

Developer Tools: Github/GitLab actions, DockerHub, DigitalOcean, Pytest, VS Code, Visual Studio, PyCharm, Jupyter notebook, Google colab

Libraries: Spice Toolkit, NumPy, Astropy, Poliastro, Pandas, Scikit, Matplotlib, SkyField, Tensorflow/Keras, OpenCV

Firmware/Hardware: Ardupilot, PX4, Betaflight, Navio2, Raspberry Pi 4, Jetson Nano, Intel NUC, CompactRIO